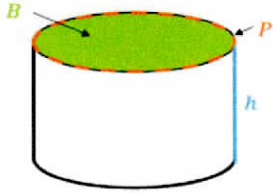


Unit 5: Solving Equations

1. The formula for the total surface area of a cylinder, S is $S = 2B + Ph$, where B is the area of the base, P is the perimeter of the base, and h is the height of the cylinder. Rearrange the formula for the surface area of a cylinder to isolate the variable B



$$\begin{aligned}
 S &= 2B + Ph \\
 -Ph &\quad -Ph \\
 \hline
 S - Ph &= 2B \\
 \frac{S - Ph}{2} &= \frac{2B}{2} \\
 \hline
 \frac{S - Ph}{2} &= B
 \end{aligned}$$

2. Tyler buys chia seeds and oats at a local store. Last month, he purchased 0.80 pounds of chia seeds and 2.5 pounds of oats for \$4.96. This month, he bought 0.40 pounds of chia seeds and 4 pounds of oats for \$5.78. How much is each food per pound?

$$\begin{aligned}
 .8c + 2.5o &= 4.96 \\
 -2(.4c + 4o &= 5.78) \rightarrow +.8c - 8o = -11.56 \\
 \hline
 -5.5o &= -6.6 \\
 \frac{-5.5}{-5.5} &\quad \frac{-6.6}{-5.5} \\
 \hline
 o &= 1.2
 \end{aligned}$$

$$\begin{aligned}
 .8c + 2.5(1.2) &= 4.96 \\
 .8c + 3 &= 4.96 \\
 -3 &\quad -3 \\
 \hline
 .8c &= 1.96 \\
 \frac{.8c}{.8} &\quad \frac{1.96}{.8} \\
 c &= 2.45
 \end{aligned}$$

3. Mrs. Smith is arranging a field trip to the MOSH museum. The budget for the field trip is \$900. The ticket price is \$9 and the cost of renting a bus is \$200. There will be 12 chaperones. Identify all the parameters in the given situation.

$$\text{Budget} = \leq 900$$

$$\text{Costs} = 9t + 200$$